# Circulating protein biomarkers for Ischaemic Stroke diagnosis

Submitted by

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In partial fulfilment of the

requirements for the Degree of

Bachelor of Engineering (Biomedical Engineering)

National University of Singapore

Session 2025/2026

**Summary**

No diagnostic tool currently exists that is fast, accessible, and accurate enough to meaningfully accelerate ischaemic stroke triage before neuroimaging becomes available, despite biological signals entering systemic circulation within minutes of arterial occlusion. The central obstacle is methodological: existing multi-biomarker studies are fragmented in design, inconsistent in normalisation, and rarely grounded in the pathophysiological mechanisms that justify marker inclusion.

This study proposes a unified diagnostic pipeline in which biological knowledge, normalisation strategy, and machine learning validation are treated as explicitly interdependent rather than isolated choices, applied to a plasma protein panel quantified from 30 clinical samples.

The findings demonstrate that a biologically informed and computationally integrated approach can extract meaningful and interpretable diagnostic signals from a small biomarker panel, directly validating the multi-biomarker panel design.

While the framework shows clear proof-of-concept promise, several methodological constraints currently limit its generalisability. The conditions under which this work can be meaningfully advanced are well-defined and form the basis of the recommendations presented at the close of this thesis.

**Acknowledgement**

The author would like to express sincere gratitude to Professor Shao Hui Lin and Tan Hong Min for their invaluable guidance, constructive feedback, and unwavering support throughout the course of this project. The author also extends appreciation to Auginia Natalia for her guidance and assistance during the laboratory work. Finally, the author gratefully acknowledges the Department of Biomedical Engineering at the National University of Singapore for providing the computational resources necessary to carry out this research.

Table of Contents

Summary ..... 2

Acknowledgement ..... 3

List of Figures ..... 6

List of Tables ..... 7

List of Symbols ..... 8

1. Introduction ..... 10

1.1. Ischaemic Stroke Burden ..... 10

1.2. Pathophysiology of IS ..... 11

1.3. Current Methods ..... 12

1.4. Circulating Biomarkers in Stroke Diagnosis ..... 13

1.5. Machine Learning in Biomedical Diagnosis ..... 13

1.6. Study Objectives & Aims ..... 14

2. Methods ..... 16

2.1. Study Design Overview ..... 16

2.2. Biomarker Selection ..... 16

2.3. Study Cohort ..... 19

2.4. Biomarker Quantification ..... 19

2.5. Data Preprocessing ..... 20

2.6. Exploratory data visualization ..... 21

2.7. Machine Learning Models ..... 22

2.8. Validation Strategy ..... 23

2.9. Model Evaluation ..... 24

2.10. Feature Importance Analysis ..... 24

2.11. Statistical Analysis ..... 25

3. Result ..... 27

3.1. Exploratory Data Visualization ..... 27

3.2. Machine Learning Model Performance ..... 28

3.3. Feature Importance Analysis ..... 31

4. Discussion ..... 35

4.1. Summary of key findings ..... 35

4.2. Biological Interpretation of Key Biomarkers ..... 36

4.3. Comparison with Existing Literature ..... 37

4.4. Clinical Implications ..... 40

4.5. Limitations ..... 41

Conclusion ..... 44

Recommendation & Future work ..... 45

References ..... 47

Appendix ..... 55

8.1. Appendix A.1 - Exploratory Data Visualisation ..... 55

8.2. Appendix A.2 - Machine Learning Classification Pipeline ..... 56

8.3. Appendix A.3 - Statistical Analysis ..... 57

**List of Figures**


1. Biomarker framework of 9-marker panel into four stroke-related biological axes
2. Illustration of Chemiluminescent ELISA principle for biomarker quantification
3. (a) Patient-level heatmap of normalised biomarker concentrations; (b) Pearson correlation matrix of the 9-marker panel
4. (a) LOOCV classification results for LR, RF, and GB
4. (b) Comparison of key evaluation metrics across models; (c) ROC curves across models
5. (a) Biomarker importance across models within LOOCV folds
5. (b) Post-hoc biomarker importance across the entire dataset
5. (c) Biomarker-level significance testing with nominal and Bonferroni-corrected thresholds
6. Python code for generating the Patient level Biomarker heatmap (Appendix A.1)
7. Python code for computing and visualising the Pearson correlation matrix (Appendix A.1)
8. Python code for the full machine learning classification pipeline (Appendix A.2)
9. Python code for the Mann-Whitney U test across all nine biomarkers (Appendix A.3)

## List of Tables

1. Table of Representative circulating biomarkers of Ischaemic Stroke spanning four biological axes

## List of Symbols

| | |
|---|---|
| **AUC** | Area under the receiver operating characteristic curve |
| **BBB** | Blood-brain barrier |
| **BSA** | Bovine serum albumin |
| **CD9** | Cluster of differentiation 9 |
| **CD63** | Cluster of differentiation 63 |
| **CRP** | C-reactive protein |
| **CT** | Computed Tomography |
| **CV%** | Coefficient of variation |
| **ELISA** | Enzyme-linked immunosorbent assay |
| **EV** | Extracellular vesicles |
| **Fg** | Fibrinogen |
| **Fn** | Fibronectin |
| **GB** | Gradient Boosting |
| **GFAP** | Glial fibrillary acidic protein |
| **HRP** | Horseradish peroxidase |
| **IgG** | Immunoglobulin G |
| **IS** | Ischaemic stroke |
| **LOOCV** | Leave-One-Out Cross-Validation |
| **LR** | Logistic Regression |
| **MRI** | Magnetic resonance imaging |

| | |
|---|---|
| **MWU** | Mann–Whitney U |
| **NfL** | Neurofilament light chain |
| **PBS** | Phosphate-buffered saline |
| **RF** | Random Forest |
| **RLU** | Relative Light Unit |
| **ROC** | Receiver operating characteristic |
| **SNR** | Signal-to-noise ratio |
| **Tau13** | Total tau protein |
| **TIA** | Transient Ischaemic Attack |
| **vWf** | von Willebrand factor |
| **α2M** | Alpha-2-macroglobulin |

# 1. Introduction

## 1.1. Ischaemic Stroke Burden

In Ischaemic stroke (IS), time is critical: approximately 1.9 million neurons are irreversibly lost for every minute of delayed reperfusion [1], creating a biological time constraint that current diagnostic approaches are often too slow to overcome. Stroke is a leading cause of mortality and long-term disability worldwide, with IS accounting for 65.3% of all cases [2]. In 2021, an estimated 7.8 million new IS cases were reported globally, with an age-adjusted incidence of 92 per 100,000 population [3].

Beyond mortality, IS produces profound and sustained impairment across mobility, self-care, and psychological well-being, with worsening modified Rankin Scale scores and declining health utility values observed at both 3 and 12 months post-stroke [4] [5], indicating that a substantial proportion of survivors fail to achieve meaningful functional recovery within the first year. The burden extends to families, with average weekly informal caregiving hours increasing from 64.3 at 3 months to 76.6 at 12 months post-stroke [6]. Globally, stroke-related healthcare expenditure and productivity losses exceed US$890 billion annually [7].

These pressures are concretely illustrated in Singapore, where IS accounts for approximately 80% of all stroke cases and ranks as the fourth leading cause of death [8]. As a high-density urban setting with a rapidly ageing population and rising vascular risk burden, Singapore exemplifies the diagnostic pressures that emerge when demographic and epidemiological trends converge, and where demand for rapid, scalable, minimally invasive detection is most urgent given imaging capacity constraints during high-volume emergency situations.

Understanding the biological mechanisms underlying ischaemic injury is therefore essential, as these processes determine disease progression and provide the molecular foundation for circulating signals amenable to rapid, non-invasive measurement.

### 1.2. Pathophysiology of IS

IS is initiated by cerebral arterial occlusion, most commonly through thrombotic or embolic mechanisms, producing abrupt reduction in cerebral blood flow and rapid metabolic failure. The ischaemic region is classically divided into a necrotic core, where energy depletion is irreversible within minutes, and a surrounding penumbra of metabolically compromised but potentially salvageable neurons, the fate of which depends on the speed of reperfusion [9].

At the cellular level, ischaemia triggers interconnected injury processes that generate measurable molecular signals in systemic circulation. Neuronal and axonal breakdown releases cytoskeletal and scaffolding proteins into the extracellular space, while astrocyte activation and injury release glial-specific structural components, together constituting a circulating signature of neuroglial injury. Activation of the coagulation cascade generates protein signals associated with thrombus formation and haemostatic dysregulation, while a systemic acute-phase inflammatory response drives hepatic synthesis and release of acute-phase proteins within hours of onset [9]. In addition to passive protein release, the ischaemic environment actively stimulates secretion of extracellular vesicles (EVs) from neurons, glial cells, and endothelial cells, carrying surface markers and molecular cargo that reflect their cellular origin [10]. Collectively, these processes give rise to a multidimensional circulating signature spanning four key axes: (i) neuroglial injury, (ii) thrombosis and haemostasis, (iii) systemic inflammation, and (iv) cellular activation and signalling.

### 1.3. Current Methods

Translating this biological understanding into timely clinical diagnosis remains the central challenge. Effective treatments exist, but their benefit is contingent on administration within narrow therapeutic windows: intravenous thrombolysis within approximately 4.5 hours of onset [11] and mechanical thrombectomy within a separately defined window dependent on perfusion imaging criteria. Consequently, the primary challenge in IS management lies not in the lack of treatment options, but in achieving timely and accurate diagnosis.

Current frontline diagnostics, predominantly Computed Tomography (CT) and Magnetic Resonance Imaging (MRI), are constrained by their dependence on structural changes that manifest only after irreversible tissue injury has begun, placing them fundamentally out of step with the biological timescale of IS. Both modalities require specialised equipment and trained personnel; CT often lacks sensitivity in detecting early ischaemic changes, whilst MRI, although more sensitive, involves longer acquisition times and remains less accessible in emergency contexts. Pre-hospital screening tools such as FAST offer rapid triage but demonstrate only moderate accuracy, with pooled sensitivity of 0.77 and failure to detect up to 40% of posterior circulation strokes [12]. Similarly, no individual blood-based protein marker demonstrates sufficient diagnostic utility for clinical use [13]. These limitations are compounded by the frequently subtle, heterogeneous, and non-specific early clinical presentations of IS, contributing to missed or delayed diagnoses in patients with mild or atypical symptoms, collectively underscoring the need for diagnostic strategies that are rapid, accessible, and deployable prior to imaging.

### 1.4. Circulating Biomarkers in Stroke Diagnosis

The molecular events described in Section 1.2 begin unfolding within minutes of arterial occlusion, releasing circulating signals that precede the structural changes detectable by imaging. Blood-based biomarkers, encompassing both soluble proteins and EV-associated markers, are therefore uniquely positioned to meet this need, providing a dynamic biochemical window into an evolving pathophysiology that opens earlier than imaging, requires no specialist equipment, and is well suited for rapid and potentially point-of-care applications [14].

However, IS is inherently multifactorial, and the biological response to ischaemia spans all four injury axes, each contributing distinct yet overlapping signals to systemic circulation. Reliance on a single biomarker therefore risks both false negatives, when a pathway is not dominant, and false positives, when similar pathways are activated by other conditions. This has driven interest in multi-biomarker panels informed by biological and clinical knowledge rather than purely data-driven approaches [15]. Translation of such panels faces two key challenges, substantial biological variability due to age, comorbidities, medication use, circadian rhythms, and time since onset [16]; and technical variability introduced by inter-assay drift and operator-dependent effects. Robust normalisation strategies are therefore essential for any biomarker framework intended for broader clinical application.

### 1.5. Machine Learning in Biomedical Diagnosis

Addressing biological variability and normalisation is therefore necessary, but not sufficient. Even a well-curated and normalised multi-biomarker panel presents an analytical challenge that conventional statistical approaches are poorly equipped to address. Traditional methods assume biomarkers contribute independently and linearly; however, IS is a heterogeneous disorder in which diagnostic signal is distributed across multiple correlated markers and encoded in the

multivariate structure of the panel, not in the absolute levels of individual measures [17]. Supervised machine learning is therefore well-positioned here: tree-based ensembles and regularised classifiers can capture complex feature interactions, accommodate correlated inputs, and be optimised for clinically relevant metrics such as sensitivity [18]. Critically, these approaches yield interpretable feature importance measures, allowing the diagnostic contribution of each biomarker to be mapped back to its underlying injury axis.

However, the application of these methods to stroke biomarker research remains methodologically immature. Systematic review of the broader multiomics ML literature confirms that existing models are characterised by small sample sizes, heterogeneous designs, and inconsistent validation practices that undermine reproducibility [19], and the same failure modes are evident across individual protein- and EV-specific IS studies. A recurring issue is the absence of biologically grounded feature selection: although multi-biomarker profiling outperforms single markers diagnostically [20], underlying models are typically constructed without reference to the biological mechanisms justifying marker inclusion, limiting interpretability and generalisability. A further issue is the absence of standardised normalisation, which allows technical variation to be absorbed into models as biological signals, directly confounding external validation [21].

### 1.6. Study Objectives & Aims

This study develops a biologically grounded and computationally robust diagnostic framework for IS, integrating biomarker selection, data normalisation, and supervised machine learning into a unified and interpretable pipeline. It is structured into three interconnected stages:

(i) Systematic identification and assembly of plasma protein biomarkers into a multi-marker panel capturing each of the four injury axes,

(ii) Development and evaluation of supervised machine learning classifiers on normalised biomarker data, with robustness assessed through Leave-One-Out Cross Validation (LOOCV) and performance quantified using clinically relevant metrics, and

(iii) Feature importance analysis identifying the biomarkers most strongly contributing to disease discrimination, interpreted within the context of their pathophysiological axes.

The central contribution of this work is to demonstrate, at proof-of-concept scale, that treating biological knowledge, normalisation strategy, and model validation as explicitly interdependent components of a unified pipeline produces a more interpretable and internally consistent diagnostic framework than the fragmented approaches that currently dominate literature.

# 2. Methods

## 2.1. Study Design Overview

This study was designed as a stepwise, biomarker-driven computational framework for the classification of hyperacute IS, structured across three sequential aims.

**Aim 1** covered biomarker selection and quantification, assembling a nine-marker plasma protein panel through a chemiluminescent Enzyme-linked immunosorbent assay (ELISA) workflow and applying signal-to-noise normalisation against an Immunoglobulin G (IgG) isotype control to account for non-specific antibody binding.

**Aim 2** covered model development and evaluation, training three supervised machine learning classifiers on z-score standardised biomarker data and assessing performance under LOOCV using confusion matrix analysis, Receiver operating characteristic (ROC) curve analysis, and Area under the receiver operating characteristic curve (AUC).

**Aim 3** covered biological interpretation, applying permutation-based feature importance analysis within the LOOCV framework to map classifier outputs back to the underlying pathophysiological axes. The following sections describe each component of this pipeline in detail.

## 2.2. Biomarker Selection

In fulfilment of Aim 1, biomarkers were selected to assemble a nine-marker plasma protein panel with systematic coverage of the four pathophysiological axes underlying IS pathophysiology: neuroglial injury, thrombosis and haemostasis, systemic inflammation, and cellular activation and signalling. The four pathophysiological axes and their representative biomarkers are illustrated in Figure 1.

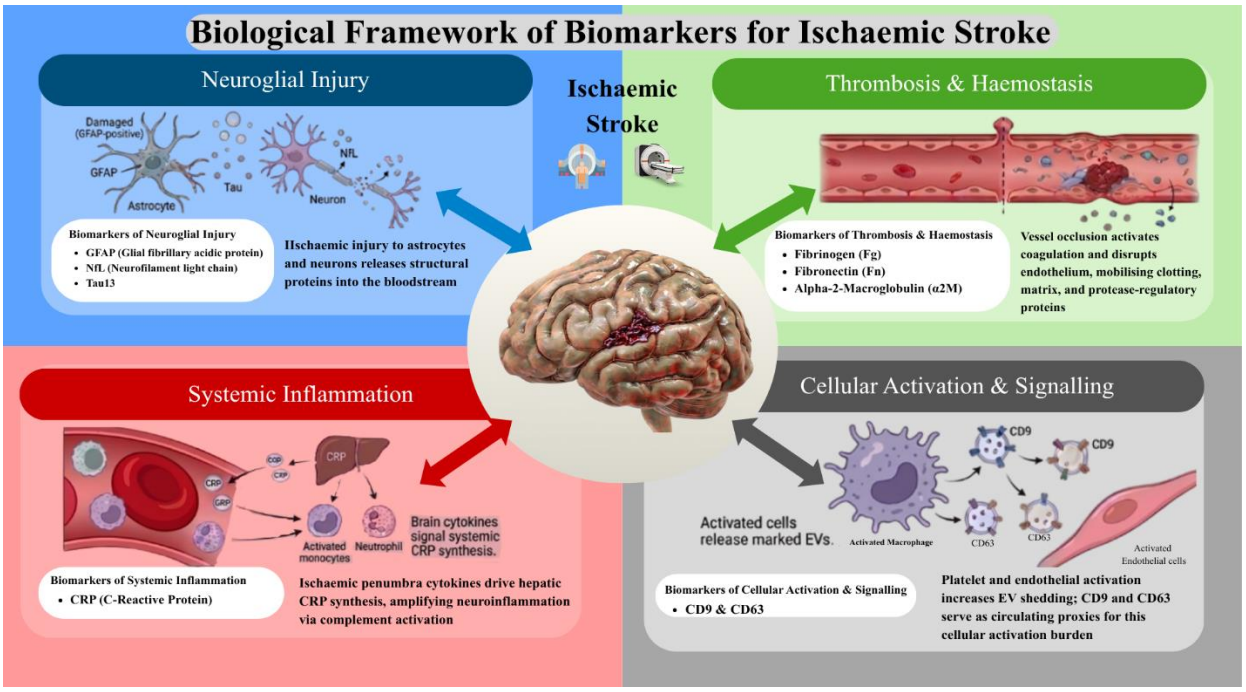



***Figure 1.*** *Four-axis framework of circulating biomarkers in ischemic stroke, capturing distributed diagnostic signals across injury pathways.*

The panel was deliberately designed to span all four axes rather than concentrate markers within a single pathway, reflecting the multidimensional nature of ischaemic injury. Biomarkers were selected based on prior research demonstrating altered plasma levels in acute IS, biological specificity to the relevant pathophysiological axis, and practical suitability for quantification using chemiluminescent ELISA. The biological basis, pathophysiological role, and supporting literature for each selected biomarker are summarised in Table 1 below.

**Table 1.** *Representative circulating biomarkers of IS span four biological axes*

| **Category** | **Representative Protein Biomarker** | **Pathophysiological Role in IS** |
|---|---|---|
| Thrombosis & haemostasis markers | **Fg**, **Fn**, **α2M** | **Fg** and **Fn** support clot formation**;** **α2M** modulates proteolytic activity in coagulation and fibrinolysis. [22][23][24] |
| Neuroglial injury markers | **GFAP**, **NfL**, **Tau13** | **GFAP** reflects astroglial activation and blood-brain barrier (BBB) disruption; **NfL** and **Tau13** (Detected using the Tau13 antibody clone) indicate axonal and neuronal injury. [25][26][27] |
| Inflammation & acute-phase response | **CRP** | **CRP** is the most extensively validated acute-phase marker in IS, with consistent evidence of early elevation post-onset [28][29], and was therefore selected as the sole representative of this axis. |
| Cellular Activation & Signalling | **CD9**, **CD63** | Although **CD9** and **CD63** are membrane-bound tetraspanins, their plasma levels reflect active vesicle release associated with vascular activation and inflammatory signalling, consistent with the adopted biomarker framework. [30] |

Having established the biological rationale for the selected panel, the following section describes the clinical samples against which these biomarkers were quantified.

### 2.3. Study Cohort

Thirty anonymised plasma samples were obtained, comprising 15 patients with a confirmed clinical diagnosis of IS and 15 healthy controls with no prior history of stroke or cerebrovascular disease. IS samples were collected during the acute phase of illness, capturing the early post-onset window in which circulating biomarker signals are most diagnostically relevant. As samples were provided in anonymised form, granular clinical metadata, including age, sex, comorbidity profile, and precise time from symptom onset to collection, were unavailable. The acute-phase designation reflects the collection protocol of the originating clinical team rather than independently verified onset-to-draw intervals.

All individuals were recruited in accordance with IRB-approved protocols, and written informed consent was obtained from each individual, or their authorised representative, prior to sample collection. Samples were subsequently provided to the research team in fully anonymised form, precluding re-identification of any participant.

### 2.4. Biomarker Quantification

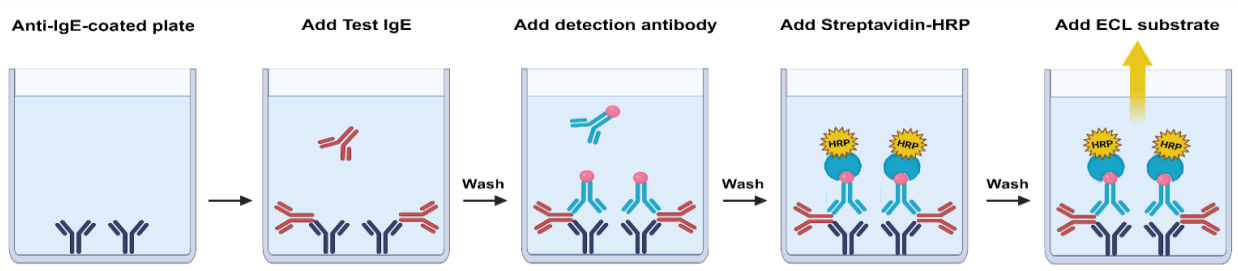



**Figure 2:** *Principle of the Chemiluminescent ELISA assay. Reproduced from BPS Bioscience*

Non-specific binding was blocked using 2% BSA in PBS (200 µL, 2 hours, room temperature), after which wells were washed with PBST (PBS containing 0.05% Tween-20). Detection was

carried out by sequential incubation with biotinylated primary antibodies (1 µg/mL in 1% BSA/PBS, 100 µL, 2 hours, room temperature) and streptavidin-HRP (1:10,000 in 1% BSA/PBS, 100 µL, 1 hour, room temperature), with PBST washes between each step to minimise non-specific retention.

Chemiluminescent signal was generated by adding 100 µL of SuperSignal ELISA Pico working solution (equal volumes of Luminol/Enhancer and Stable Peroxide Solution) to each well, and Relative Light Unit (RLU) were measured using a luminometer within 1 to 5 minutes of substrate addition. All samples were run in technical triplicate alongside an IgG isotype control, with each condition measured under identical assay parameters. The triplicate IgG readings were averaged to yield a single mean background RLU value, against which each individual sample replicate was divided to apply signal-to-noise normalisation. The three resulting normalised values were averaged to produce a single consolidated value per sample per biomarker, which was subsequently passed to the data processing pipeline described in the following section.

### 2.5. Data Preprocessing

Signal-to-noise normalisation was applied at the individual replicate level prior to averaging, as described in Section 2.4. The resulting Signal/Noise values were structured into a data matrix, with each row representing an individual plasma sample and each column representing the averaged normalised value for a given biomarker. Class labels were assigned as binary variables, with healthy controls denoted as '0' and IS patients denoted as '1'.

A CV% threshold of 30% was applied to assess technical reproducibility of triplicate measurements, reflecting the inherently higher signal variability associated with chemiluminescent detection relative to conventional colorimetric ELISA formats, for which the standard 20%

threshold is typically cited. All triplicate measurements across the dataset fell within this threshold, confirming acceptable reproducibility across all samples and biomarkers.

To improve visual interpretability, additional preprocessing steps were applied to the data matrix before exploratory visualisation. A log(1 + x) transformation was applied using NumPy's log1p function to reduce skewness and stabilise variance across biomarker distributions, followed by global min-max normalisation to scale all values to a common range of 0 to 1, enabling comparative visualisation across biomarkers with differing signal magnitudes. These transformation steps were applied exclusively for visualisation purposes; the raw S/N normalised values served as direct input to model training.

Before model fitting within each LOOCV fold, features were standardised to zero mean and unit variance using z-score scaling. The scaler was fitted exclusively on the $n - 1$ training samples and the derived parameters subsequently applied to the held-out sample, ensuring no information from the test observation informed the scaling transformation.

### 2.6. Exploratory data visualization

Exploratory data visualisation was conducted to assess biomarker signal distributions, examine group-level separation between IS and healthy control samples, and detect any anomalous patterns that could confound downstream modelling. Two complementary visualisations were generated using Seaborn and Matplotlib.

A patient-level heatmap was generated from the biomarker matrix following log(1 + x) transformation and global min-max normalisation. These steps were applied to improve colour distinguishability across the wide dynamic range of the data, and were used exclusively for visualisation, providing an overview of relative biomarker signal intensities across all samples simultaneously, enabling visual inspection of coordinated expression patterns, multi-marker

signatures, and potential group-level differences. IS and healthy control samples were arranged in separate blocks to facilitate direct visual comparison between groups.

A Pearson correlation matrix was computed directly using pandas and Seaborn from the signal-to-noise normalised values without further rescaling, and visualised as an annotated heatmap. As Pearson correlation captures linear relationships between continuous variables rather than absolute signal magnitudes, additional rescaling is unnecessary. This approach assessed whether biomarkers within and across the four pathophysiological axes were responding independently or in a coordinated fashion, and identified any potential redundancy within the panel that could inform interpretation of machine learning outputs.

### 2.7. Machine Learning Models

In fulfilment of Aim 2, three supervised classifiers were trained on z-score standardised biomarker data and evaluated under LOOCV using confusion matrix analysis, ROC curve analysis, and AUC. Classifiers were selected to span a spectrum of model complexity, from an interpretable linear baseline to nonlinear ensemble approaches, enabling systematic comparison of linear and nonlinear representations of IS pathophysiology.

Logistic Regression (LR) was included as an interpretable linear baseline, implemented with the liblinear solver, a maximum of 10,000 iterations to ensure convergence, and class_weight='balanced'. Its linear formulation provides a biologically transparent reference against which the nonlinear models can be compared.

Random Forest (RF) was selected for its ability to capture higher-order biomarker interactions through ensemble averaging, implemented with 100 estimators and class_weight='balanced'; its variance-reducing properties make it well suited to small, feature-rich datasets.

Gradient Boost (GB) was included as a sequential ensemble method that constructs trees iteratively, with each successive tree trained to correct the residual errors of the ensemble thus far. This iterative error correction allows GB to model complex, non-additive relationships that RF may average out. It was implemented with 100 estimators, a maximum tree depth of 3 to constrain model complexity, and a learning rate of 0.1. As GradientBoostingClassifier does not natively support class_weight, balanced sample weights were passed via compute_sample_weight('balanced') directly to the fit() call, consistent with the balancing strategy applied across all classifiers. A fixed random seed of 42 was applied to all stochastic components of the pipeline to ensure full reproducibility.

### 2.8. Validation Strategy

To ensure robust and unbiased evaluation of model performance, LOOCV was implemented as the primary validation strategy. Given the limited sample size of 30, LOOCV was selected as it maximises the use of available data for training while providing an independent test observation at each iteration: at each fold, a single sample is withheld as the test instance and the remaining $n - 1$ samples are used for training. Within each fold, z-score scaling was applied exclusively to the training subset, with the derived parameters subsequently applied to the held-out sample, preventing data leakage. Predictions from all iterations were aggregated to compute overall performance metrics.

Whilst LOOCV can produce higher-variance estimates than strategies using larger test partitions, and is sensitive to outlier samples at small n, a fixed train–test split at this cohort size would substantially reduce training data, increasing the risk of underfitting and unstable model estimates, making these trade-offs acceptable in context. The variance and outlier sensitivity implications of LOOCV are addressed in detail in Section 4.5.

### 2.9. Model Evaluation

Classifier performance was quantified using a complementary set of metrics: confusion matrix, accuracy, sensitivity, specificity, and AUC. Relying on a single aggregate measure is insufficient for this dataset, where a balanced but small cohort of 30 samples means that class-specific error patterns, particularly missed IS diagnoses, may be obscured by overall correctness alone.

The confusion matrix decomposes outcomes into true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN), enabling identification of systematic misclassification patterns. Accuracy [(TP + TN) / N], sensitivity [TP / (TP + FN)], and specificity [TN / (TN + FP)] were derived from these counts and interpreted jointly rather than in isolation, with sensitivity accorded particular clinical weight given that false negatives correspond to missed IS diagnoses in a time-critical setting.

The AUC summarises classifier performance across all decision thresholds by characterising the sensitivity–specificity trade-off, ranging from 0.5 (chance) to 1.0 (perfect discrimination), and serves as the primary metric for cross-classifier comparison. All metrics were computed on aggregated held-out LOOCV predictions, ensuring unbiased estimates of generalisation performance. To quantify estimation uncertainty around the AUC, bootstrapped 95% confidence intervals were computed via 1,000 resamples of the aggregated LOOCV prediction vector, with resamples yielding a single class excluded as AUC is undefined in such cases.

### 2.10. Feature Importance Analysis

Feature importance analysis was conducted to translate predictive performance into biologically interpretable signals, in accordance with Aim 3. Permutation importance was selected for its model-agnostic properties, enabling consistent comparison across classifiers with differing internal structures. The approach operates by randomly permuting each feature and measuring the

resulting decrease in performance; larger decreases indicate greater feature relevance to the classifier's decision boundary.

Importance was estimated within each LOOCV fold using only the training subset, implemented via scikit-learn's permutation importance with 10 repeats per fold, balancing estimation stability against computational cost across all 30 folds. Raw scores were clipped at zero to suppress noise-induced negative values, normalised to sum to one within each model using a small epsilon ($1e{-8}$) for numerical stability, and averaged across folds to yield stable per-feature estimates. Features were ranked in descending order of mean importance.

A secondary analysis was conducted by refitting each model on the full dataset using 30 permutation repeats, serving as a stability check on whether feature rankings remained directionally consistent under increased estimation precision. As these full-dataset estimates derive from training data, they carry no generalisation claim and are interpreted solely as a cross-check against the LOOCV-derived rankings.

Importance profiles were subsequently mapped onto the four pathophysiological axes of IS established in the Introduction, contextualising model-derived rankings within established mechanisms of ischaemic injury. Given that permutation importance reflects association rather than causation, these findings are treated as hypothesis-generating and subject to further validation.

### 2.11. Statistical Analysis

To complement the machine learning evaluation with statistical evidence, a Mann–Whitney U (MWU) test was applied independently to each of the nine biomarkers, comparing signal-to-noise–normalised values between IS patients and healthy controls. The MWU test was selected as a non-parametric alternative, as normality cannot be reliably assessed at $n = 30$ and biological outliers

are expected in plasma biomarker data. A two-sided alternative was specified, as no directional hypothesis was imposed regarding whether biomarker levels would be elevated or reduced in IS. Given that nine independent tests were conducted, a Bonferroni correction was applied, yielding an adjusted significance threshold of $p < 0.0056$. Whilst conservative, this controls the family-wise error rate at the cost of increased type II error; results were therefore interpreted in conjunction with model-derived feature importance rather than in isolation. Effect sizes were quantified using rank-biserial correlation to characterise the magnitude of group differences independent of sample size.

# 3. Result

## 3.1. Exploratory Data Visualization

To characterise biomarker distributions and assess group separation prior to classification, exploratory visualisation was performed on the quantified nine-marker panel.

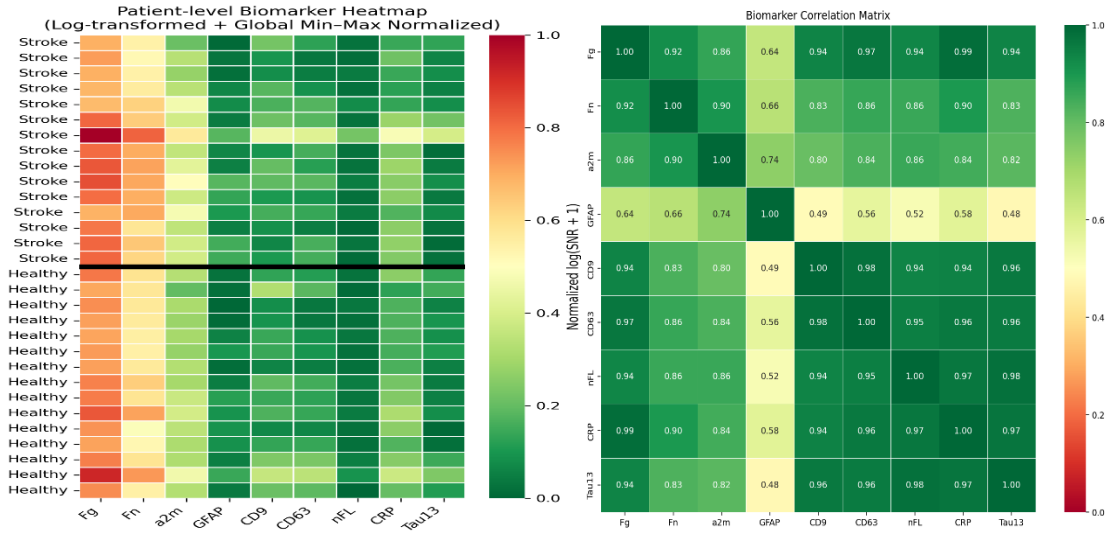



**Figure 3.** *(a) Patient-level heatmap of normalized biomarker concentrations. (b) Pearson correlation matrix of the nine-marker panel; colour intensity denotes strength.*

**Figure 3a** reveals heterogeneous distributional profiles across the panel. Thrombosis and haemostasis markers showed the most pronounced group differences: **Fg** and **Fn** were predominantly elevated in IS samples relative to healthy controls, though outliers in opposing directions tempered the apparent separation. **α2M** displayed a subtler upward shift in IS with substantial inter-group overlap. **CRP** showed moderate, largely overlapping distributions across both groups. Neuroglial markers were more variable; **GFAP** demonstrated high intra-group variability and considerable inter-group overlap with healthy controls, whilst **NfL** and **Tau13** were uniformly low across both groups with isolated IS outliers. **CD9** and **CD63** were generally low

across the full cohort, with modest elevations confined to a subset of IS samples rather than a consistent group-wide shift.

**Figure 3b** reveals strong positive intercorrelations among the majority of biomarkers (r = 0.80–0.99), indicating considerable co-expression across the coagulation, inflammatory, and neurological axes. **GFAP** was a notable exception, exhibiting markedly weaker correlations with all other biomarkers (r = 0.48–0.74), consistent with its role as a glial-specific structural protein mechanistically distinct from the systemic processes reflected by markers such as **Fg**, **CRP**, and **Fn**. The pervasive intercorrelations among the remaining biomarkers suggested potential multicollinearity, reinforcing the selection of ensemble classifiers robust to correlated feature spaces and the use of permutation importance to identify the most discriminative signals rather than treating individual biomarker values in isolation.

Collectively, no single biomarker afforded reliable group separation, and pervasive intercorrelations precluded simple univariate discrimination, motivating the application of supervised machine learning to integrate multivariate biomarker information and model complex feature interactions.

### 3.2. Machine Learning Model Performance

To address these limitations, three supervised classifiers were evaluated on their ability to discriminate IS from healthy cases using a nine-biomarker panel. Given the limited sample size of 30 participants, LOOCV was employed to maximise data utility whilst providing an unbiased estimate of generalisation performance.

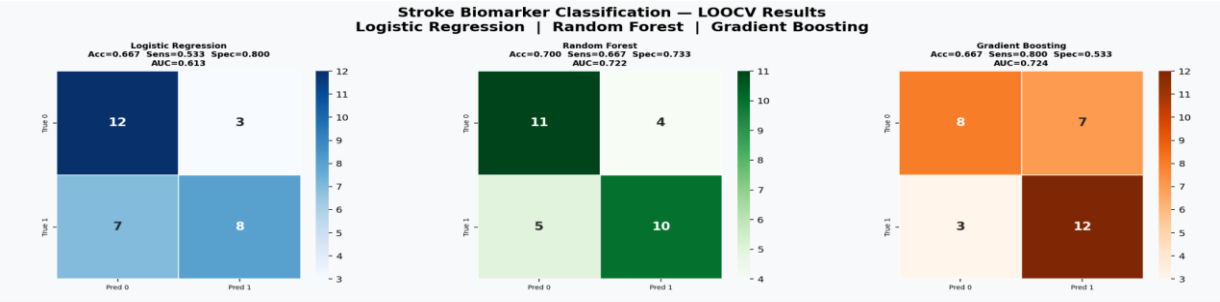
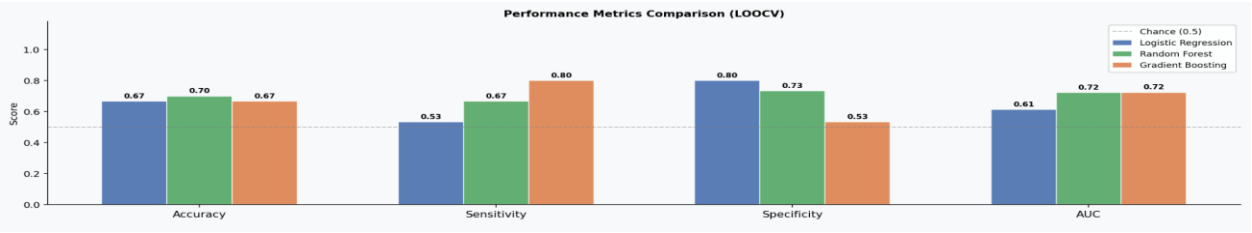



**Figure 4.** *(a) Confusion matrices for LR, RF, and GB models. Matrices display predicted vs. true labels for IS (Class 1) and healthy controls (Class 0).*

The aggregated LOOCV confusion matrices are presented in **Figure 4a**. LR correctly classified 12 healthy and 8 IS cases, yielding 3 false positives and 7 false negatives. RF correctly identified 11 healthy and 10 IS cases, with 4 false positives and 5 false negatives. GB produced the most asymmetric error profile, with 8 healthy and 12 IS, 7 false positives, and only 3 false negatives. In a clinical screening context, false negatives carry considerably greater consequences than false positives, as missed stroke diagnoses risk delayed intervention and poorer patient outcomes. On this basis, GB demonstrated the most clinically favourable error distribution by minimising missed stroke cases.



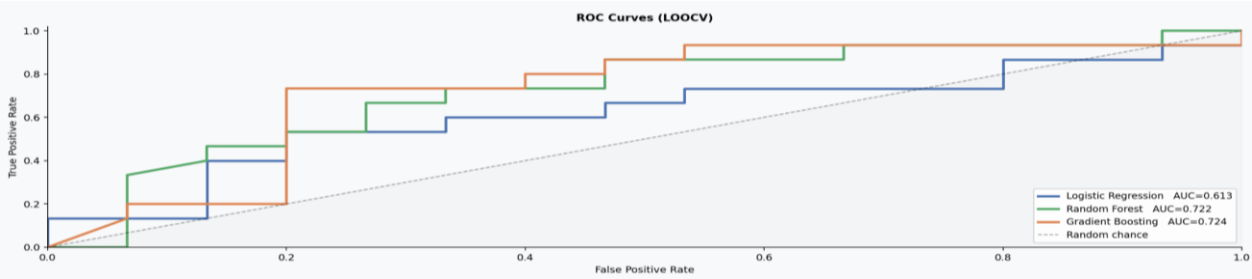



**Figure 4.** *(b) Comparison of accuracy, sensitivity, specificity, and AUC across models; (c) ROC curves for LR, RF, and GB; dashed diagonal represents random chance.*

The four-evaluation metrics are summarised in **Figure 4b**. RF achieved the highest accuracy (0.700), GB the highest sensitivity (0.800), and LR the highest specificity (0.800). The divergence between sensitivity and specificity across models reflects a fundamental trade-off inherent to threshold-based classification, whereby lowering the decision boundary increases stroke detection at the cost of more false positives, whilst raising it improves specificity at the expense of missed cases. GB's aggressive stroke detection yielded high sensitivity at the cost of reduced specificity (0.533), whilst LR's conservative decision boundary produced the inverse pattern. RF achieved the most balanced trade-off overall, with a sensitivity of 0.667 and specificity of 0.733. It is important to note that accuracy alone is insufficient as a performance measure in this context, as a naive majority classifier would achieve 50% on this balanced dataset. The superior AUC values of RF and GB therefore indicate that both models possess meaningfully greater discriminative capacity across all operating thresholds, independent of any fixed classification boundary.

The ROC curves are presented in Figure 4c. GB achieved the highest AUC (0.724, 95% CI: 0.489–0.912), followed closely by RF (0.722, 95% CI: 0.511–0.903), with LR demonstrating substantially weaker discrimination (0.613, 95% CI: 0.394–0.819). The wide and overlapping confidence intervals between GB and RF indicate that the two ensemble models are not statistically distinguishable at this sample size; GB's preference over RF is therefore grounded in its clinically superior sensitivity profile rather than aggregate discriminative superiority. That LR's interval, whilst similarly wide, sits entirely below those of both ensemble models at its lower bound provides modest directional support for its relative inferiority, though this too should be interpreted cautiously given the imprecision inherent at n=30. Of particular note, GB rises steeply to a true

positive rate of approximately 0.73 at a false positive rate of 0.20 or below, indicating that it maintains strong sensitivity even under tightened specificity constraints. This early dominance in the upper-left region of the ROC space is a clinically desirable property for a screening tool where minimising missed diagnoses is paramount.

Taken together, the confusion matrix, performance metrics, and ROC analysis consistently support GB and RF as the stronger classifiers, with GB marginally preferred on clinical grounds. These findings provide a basis for further interrogation of the relative contribution of individual biomarkers to model classification decisions.

### 3.3. Feature Importance Analysis

To address Aim 3, permutation importance was computed within the LOOCV training folds for each model to identify which biomarkers drove classification decisions, ensuring that feature relevance estimates were not contaminated by held-out test observations. Scores were normalised across all nine biomarkers within each model so that values reflect each biomarker's proportional contribution to classification and are directly comparable across models. Cross-model consistency is the primary interpretive criterion adopted here: biomarkers appearing in the top three across all three classifiers are treated as robustly ranked, whilst secondary rankings are considered indicative only. The resulting profiles are presented in **Figure 5a**.

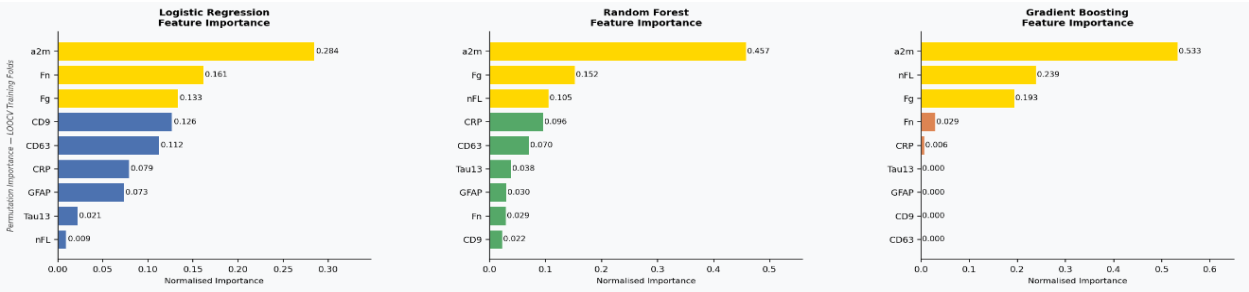



**Figure 5a.** *LOOCV permutation importance profiles for LR (blue), RF (green), and GB (orange). Yellow bars highlight the top three biomarkers per model.*

Across all three classifiers, **α2M** emerged as the dominant feature, with normalised importance scores of 0.284, 0.457, and 0.533 in LR, RF, and GB respectively. Secondary rankings diverged across models: LR ranked **Fn** (0.161) and **Fg** (0.133) second and third; RF ranked **Fg** (0.152) and **NfL** (0.105); and GB produced the most concentrated profile, with **NfL** (0.239) and **Fg** (0.193) as secondary contributors. It should be noted that permutation importance estimates at n=30 carry substantial variance, and the precise numerical ordering of secondary and lower-ranked biomarkers should not be treated as stable. What is interpretable is the cross-model pattern rather than any individual score. On this basis, the consistent appearance of **Fg** across all three models as a top three contributor is meaningful; the specific rankings among the remaining biomarkers are treated as indicative only. **CD9**, **CD63**, **GFAP**, and **Tau13** contributed negligibly across all models, a finding robust enough at this magnitude of difference to be interpreted with greater confidence than the finer distinctions among mid-ranked features.

To verify that these rankings were not an artifact of training fold subsampling, permutation importance was additionally computed on the full 30-sample dataset as a post-hoc cross-check. This does not constitute independent validation, but serves to confirm the directional consistency of the LOOCV-derived estimates. Results are presented in **Figure 5b**.

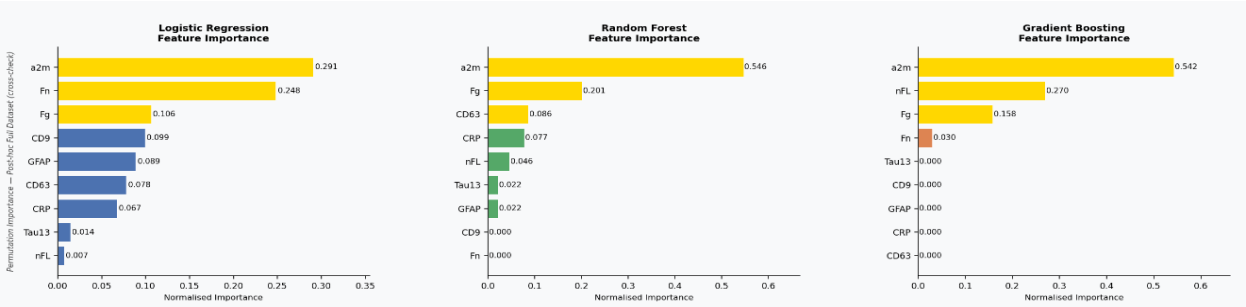



**Figure 5b.** *Full-dataset permutation importance profiles for LR (blue), RF (green), and GB (orange). Yellow bars highlight the top three biomarkers per model.*

The post-hoc rankings were largely concordant with the LOOCV findings. **α2M** retained its dominant position across all models, with scores of 0.291, 0.546, and 0.542 for LR, RF, and GB respectively. LR's secondary rankings were unchanged, with **Fn** (0.248) and **Fg** (0.106) again placing second and third. RF retained Fg (0.201) in second position, though **CD63** (0.086) replaced **NfL** in third, representing a minor divergence. GB showed near-identical ordering, with **NfL** (0.270) and **Fg** (0.158) as secondary contributors. The modest shifts observed in lower-ranked biomarkers are expected given the increased sample coverage, and do not alter the overall importance hierarchy. The consistency of **α2M** 's dominant ranking across structurally distinct models and across both analytical approaches supports the conclusion that its discriminative signal is genuine and reproducible rather than an artefact of model inductive bias or training fold subsampling.

To further contextualise the feature importance findings, an MWU test was conducted for each biomarker independently, testing whether its distribution differed significantly between the IS and healthy groups. As a non-parametric test, MWU requires no assumption of normality, making it appropriate for the small sample size. Crucially, this analysis evaluates each biomarker in isolation rather than within the multi-biomarker panel, and should therefore be interpreted as a complementary univariate characterisation rather than a direct measure of model-relevant importance. A Bonferroni correction was applied across nine comparisons, yielding an adjusted significance threshold of $p = 0.0056$. Results are presented in **Figure 5c**.

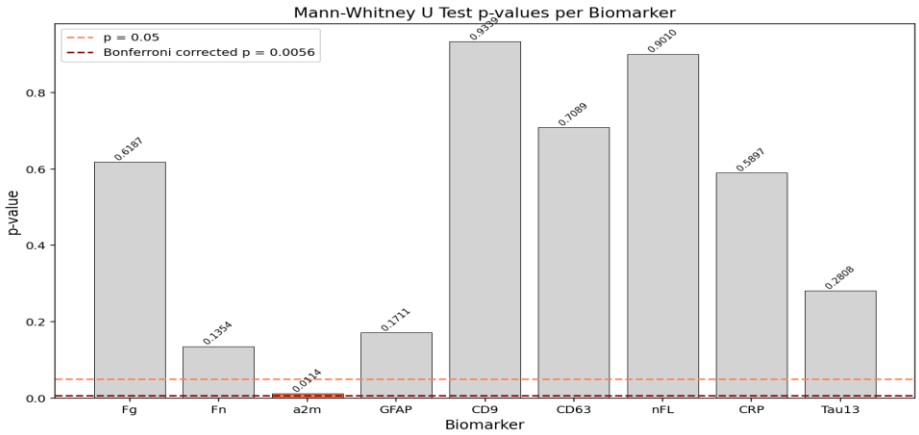



**Figure 5c.** *MWU test results per biomarker. P-values for the nine-marker panel. Dashed lines denote nominal significance and Bonferroni-corrected threshold.*

Of the nine biomarkers, only **α2M** reached conventional statistical significance (p = 0.0114), falling below the p = 0.05 threshold whilst approaching but not meeting the Bonferroni-corrected threshold. All remaining biomarkers were non-significant, with p-values ranging from 0.135 for **Fn** to 0.934 for **CD9**. This finding reinforces the rationale for a multi-biomarker panel approach, as individual biomarkers lacking univariate significance can nonetheless contribute complementary signals when modelled jointly. The alignment of **α2M** 's statistical significance on MWU testing with its dominant feature importance across all three classifiers provides corroborating support for its position as the most informative single biomarker within this panel. It should be noted that both analyses draw on the same 30 samples and are therefore not independent lines of evidence; their agreement strengthens directionality but does not constitute the degree of independent corroboration that a held-out external cohort would provide. The mechanistic relevance of **α2M** alongside that of the secondary contributors is examined in the Discussion in the context of IS pathophysiology.

# 4. Discussion

## 4.1. Summary of key findings

This study evaluated a biologically grounded nine-marker plasma protein panel for ML-based IS classification across three supervised classifiers, yielding three principal findings.

First, ensemble methods substantially outperformed the linear baseline: RF and GB achieved AUC values of 0.722 and 0.724 respectively against LR's 0.613, demonstrating that the discriminative structure encoded in this panel is inherently nonlinear.

Second, classifier selection carries consequences beyond aggregate discrimination: GB's sensitivity of 0.800 versus LR's 0.533 positions it as the more appropriate architecture for a screening context where undetected IS represents the higher-consequence error class.

Third, **α2M** emerged as the dominant discriminator across all three classifiers, with corroborating directional support from its sole achievement of conventional statistical significance on MWU testing ($p = 0.0114$); the non-independence of these two analyses, both derived from the same 30 samples, is acknowledged, and their agreement is interpreted as convergent rather than independent evidence. Most markers are individually non-significant yet collectively discriminative directly demonstrates why single-biomarker approaches are insufficient: the diagnostic signal is distributed across pathophysiological axes and is only recoverable when markers are modelled jointly.

Collectively, these findings provide proof-of-concept that a biologically informed, computationally integrated pipeline can extract meaningful IS-discriminative signals from a small plasma protein panel, whilst the performance ceiling observed under LOOCV defines the boundary conditions under which the framework must be scaled and externally validated. The identity of the

markers driving this performance, and the pathophysiological axes they represent, carries interpretive weight beyond classification metrics alone, and is examined in the following section.

### 4.2. Biological Interpretation of Key Biomarkers

The feature importance analysis consistently implicated a subset of biomarkers spanning three of the four pathophysiological axes, with **α2M** emerging as the dominant contributor across all three classifiers. Understanding why these specific markers rose to prominence requires situating them within the mechanistic biology of IS, beginning with the most consistently ranked. **α2M** is a broad-spectrum protease inhibitor and acute phase reactant that is upregulated in response to neuroinflammation and BBB disruption [31]. During IS, loss of BBB integrity triggers elevated MMP activity, which degrades tight junction proteins and basal lamina components; **α2M** counteracts this by binding and neutralising circulating proteases implicated in BBB degradation [32]. Elevated plasma **α2M** levels in stroke patients relative to healthy controls [33], provide independent biological corroboration that its dominant model-level importance reflects a genuine signal rather than a statistical artefact.

**Fg** and **Fn**, consistently ranked as secondary features across models, reflect the coagulation and vascular disruption axes of IS pathophysiology. **Fg** is a central mediator of thrombus formation, with elevated plasma levels representing both a stroke risk factor and an acute phase response to ischaemic injury [34]. **Fn** contributes to extracellular matrix integrity and has been identified as a substrate for MMP-9-mediated BBB degradation during ischaemia [35], and has been proposed as a biomarker for haemorrhagic transformation. Their co-occurrence across models suggests the coagulation and vascular disruption axis provides a discriminative signal complementary to the neuroinflammatory signal captured by **α2M**.

**NfL**, prominent in RF and GB, is a neuronal cytoskeletal protein released into peripheral blood following axonal membrane disruption [36]. It is significantly elevated in IS patients relative to healthy controls and correlates with stroke severity and functional outcome [37], reflecting the downstream neurodegeneration accompanying ischaemia and reperfusion injury. Its prominence specifically in ensemble models suggests it contributes most meaningfully through interaction with other biomarkers rather than as a standalone discriminator, consistent with its non-significant MWU result ($p = 0.901$).

The remaining biomarkers contributed minimally across models. **GFAP** and **Tau13**, whilst markers of glial activation and neurodegeneration respectively, demonstrate greater elevation in haemorrhagic stroke and chronic neurodegenerative conditions than in the acute ischaemic setting [38] [39], limiting their utility here. **CD9** and **CD63** are platelet-derived tetraspanins involved in thrombotic processes [40] but likely lack the specificity to discriminate IS from healthy controls at this sample size. **CRP**, as a non-specific inflammatory marker elevated across a broad range of pathological states, offers limited individual discriminative power [41][42].

Collectively, the feature importance profile maps onto three pathophysiological axes of IS: neuroinflammation and protease dysregulation **(α2M),** coagulation and vascular disruption **(Fg, Fn),** and structural neuronal injury **(NfL)**. The convergence of cross-validated model rankings and MWU testing around **α2M** as the primary discriminator supports the argument that a multi-biomarker panel anchored by **α2M** holds promise as a foundation for IS diagnostic development, subject to validation in larger cohorts.

### 4.3. Comparison with Existing Literature

Direct comparison with the published literature requires careful framing. Most ML-based stroke biomarker studies differ from the present work in three consequential respects: they operate at

substantially larger sample sizes, employ high-throughput platforms capable of profiling hundreds to thousands of proteins simultaneously, and several frame classifications as stroke subtype discrimination rather than IS versus healthy control differentiation. Numerical benchmarking must therefore be interpreted with these structural differences in mind.

Panel-level AUC values from large-scale plasma proteomics studies typically fall between 0.78 and 0.82 for IS-related tasks. A 2026 multiplatform proteomics study using a 61-protein panel across repeated nested cross-validation achieved an average AUC of 0.82 for distinguishing acute IS from other stroke subtypes [43], whilst a separate ML study on 456 acute IS patients reported accuracy and sensitivity both exceeding 0.73 for etiological subtype discrimination [18]. The RF and GB AUC values of 0.722 and 0.724 sit modestly below these figures, but the comparison is structurally asymmetric: both studies operated at far larger sample sizes with unbiased high-throughput discovery platforms. The present panel's nine biologically selected markers, quantified by chemiluminescent ELISA, prioritise interpretability and translational accessibility over discriminative ceiling, and the observed AUC values are consistent with what small-sample, hypothesis-driven frameworks routinely achieve; a 2018 pilot LOOCV study on a similarly sized stroke cohort reported an AUC of 0.93 whilst explicitly acknowledging optimism inflation at small n, situating the interpretive caution applied here as a recognised field-wide challenge rather than a study-specific shortcoming [44].

The most substantive comparison lies at the level of individual biomarker rankings rather than aggregate performance. The dominance of **α2M** is independently supported in the clinical literature. A study enrolled 159 IS patients and 77 controls and found serum **α2M** to be significantly elevated at admission in the IS group, with levels positively correlated with both age and white matter lesion severity [33]. The age correlation is particularly relevant here, as the

absence of demographic metadata means that some portion of **α2M**'s discriminative prominence cannot be fully disentangled from potential group-level demographic imbalance. This association is further reinforced a 2022 review that confirmed elevated **α2M** in IS patients across multiple independent cohorts [23], strengthening the case that its model-level prominence reflects genuine biological signals.

The secondary prominence of **Fg** and **Fn** is equally consistent with the literature. **Fn** appeared among externally validated protein classifiers distinguishing stroke subtypes, and **Fg**'s role as both a coagulation mediator and acute-phase reactant in IS is well-documented across multiple cohorts, reinforcing the biological plausibility of their classification contributions here [43].

Finally, the argument that panel-level modelling recovers signals that individual markers cannot is independently supported by another study, which demonstrated that individual inflammatory plasma proteins with moderate AUCs of 0.75 to 0.81 produced substantially stronger combined performance when modelled jointly, reaching an AUC of 0.96 with sensitivity of 87.5% and specificity of 96.2% [45]. This directly parallels the present finding that markers individually non-significant on MWU testing nonetheless contribute meaningfully within the ensemble, providing empirical support for the multi-biomarker panel design.

Taken together, the present findings are broadly concordant with the existing literature: the AUC range aligns with comparable small-sample frameworks, **α2M**'s dominance is independently corroborated, and panel superiority over single-marker approaches is reinforced by parallel observations. The performance gap relative to large-scale proteomics work reflects differences in cohort size and platform scope and defines the scaling conditions under which external validation would be most informative.

### 4.4. Clinical Implications

The clinical value of this framework lies not in replacing neuroimaging but in addressing the diagnostic gap that precedes it. CT and MRI remain the gold standard for confirming IS in the acute setting, but both are contingent on infrastructure, trained personnel, and patient cooperation that cannot always be guaranteed. The present panel is designed to function as an upstream of imaging, a rapid triage tool that stratifies patients by stroke likelihood before neuroimaging becomes available.

The most immediate clinical benefit is temporal. A blood draw-based screen enables earlier mobilisation of imaging resources and reduces the interval between symptom onset and treatment initiation. Given the well-established rate of neuronal loss during untreated ischaemia, even modest reductions in pre-imaging triage time carry meaningful consequences for functional outcome. This is particularly relevant in pre-hospital settings, hospitals without dedicated neuroimaging, and busy emergency departments where imaging queues routinely extend door-to-needle time beyond the thrombolysis therapeutic window.

The panel also offers practical advantages for patients in whom standard neuroimaging is contraindicated or logistically difficult. MRI is incompatible with implanted metallic devices and demands sustained stillness, which patients with motor agitation, altered consciousness, or severe distress frequently cannot provide. CT involves ionising radiation and has limited sensitivity for early ischaemic changes. Venepuncture imposes none of these constraints and is applicable across a far broader range of acute presentations.

The classifier findings carry direct deployment implications. GB's superior sensitivity positions it as the preferred architecture for triage, where the consequences of a missed IS diagnosis outweigh those of a false positive. Its false positive rate of 47% among healthy controls is acceptable within

a two-stage pathway where a positive screen triggers imaging rather than immediate treatment, and treatment decisions must therefore remain contingent on imaging confirmation. Within this architecture, GB's error profile is not a weakness, but a deliberate trade-off calibrated to the asymmetric costs of diagnostic error in acute stroke care.

These deployment considerations are particularly salient in the Singapore context introduced in Section 1.1, where high patient volumes, a rapidly ageing population, and imaging capacity constraints make pre-imaging triage capacity a genuine bottleneck. A blood draw-based screen requiring no specialist equipment is directly compatible with this setting, and GB's sensitivity-prioritising error profile aligns with the clinical imperative to minimise missed diagnoses where delayed IS intervention carries well-documented functional and economic consequences. Validation of the present framework in a Singapore-based multisite cohort would therefore serve both as an external generalisation test and as a contextually grounded proof of clinical deployability in the target setting.

Realising this clinical potential, however, requires addressing the methodological constraints that currently limit the framework's generalisability, and these are examined in the following section.

### 4.5. Limitations

Several methodological constraints qualify the interpretation of the present findings and must be considered when evaluating the generalisability of the proposed framework.

The most fundamental limitation is sample size. With 30 observations split evenly across two classes, the dataset is insufficient to support conclusions beyond proof-of-concept demonstration. LOOCV provides the least-biased available estimate of classification performance at this scale, though reliable AUC estimation in binary classification typically requires 50–100 observations per class and the present cohort therefore falls below the threshold at which performance estimates

stabilise; bootstrapped 95% confidence intervals were computed to characterise estimation uncertainty around the reported AUC values. The resulting intervals are wide, spanning 0.425 units for LR, 0.392 for RF, and 0.423 for GB, and substantially overlapping across all three models, confirming that the observed point estimates carry considerable imprecision at n=30 and that small numerical differences between models should not be over-interpreted. The true population AUC could plausibly vary substantially around the observed values, and this uncertainty directly motivates cohort expansion as the most pressing methodological priority. Because each training fold differs from the full dataset by only a single observation, the 30 resulting models are highly correlated, inflating variance in the aggregated performance estimate relative to what a genuinely independent cohort would provide. This is further compounded by LOOCV's sensitivity to outliers: at n = 30, a single anomalous sample exerts disproportionate influence on the fold from which it is excluded, potentially distorting model fitting in ways that would not persist at larger sample sizes. The stepped ROC curves in **Figure 4c** are a visual manifestation of this imprecision and should be read accordingly.

The complete absence of demographic and clinical metadata introduces three confounds that cannot be retrospectively resolved. Without age and sex data, demographic matching between IS and control groups cannot be verified, which matters because **α2M** levels are known to increase with age and differ by sex. Some portion of **α2M**'s discriminative prominence may therefore reflect group-level demographic imbalance rather than IS-specific biology alone. The absence of onset-to-draw intervals is equally consequential: markers such as **NfL** and **GFAP** have well-characterised temporal release kinetics, and their low feature importance rankings may reflect early sampling before peak release rather than genuine biological absence in IS. Finally, without comorbidity or medication records, biomarkers with broad physiological sensitivity, particularly

**CRP**, **Fg**, and **α2M**, cannot be screened for confounding by inflammatory or metabolic conditions unrelated to stroke, leaving open the possibility that compressed group differences in these markers partially reflect baseline elevations in controls rather than attenuated IS signals.

The assay platform introduces a further technical constraint. Signal-to-noise normalisation against an IgG isotype control adequately controls for non-specific binding within this dataset, but produces relative rather than absolute abundance values that cannot be mapped onto published reference ranges or used to define clinically actionable thresholds. External validation would require either adoption of the same normalisation framework or recalibration against absolute concentration standards, limiting the immediate portability of the present model.

Finally, the absence of an independent external validation cohort means that performance estimates reflect internal consistency rather than true generalisability. LOOCV cannot substitute for evaluation across different clinical sites, collection protocols, or more heterogeneous patient populations, and the degree to which the framework degrades under those conditions remains unknown.

These limitations define the boundary conditions of the framework rather than undermining its central findings. Each maps onto a resolvable methodological gap, and the conditions under which this work can be meaningfully advanced are well-defined. Specific recommendations are elaborated in the Recommendations and Future Work section following the Conclusion.

## Conclusion

This study set out to develop a biologically grounded and computationally integrated diagnostic framework for IS through three interconnected aims, and the findings demonstrate, at proof-of-concept scale, that each aim was successfully operationalised and that the pipeline is both feasible and internally consistent.

Aim 1 was fulfilled through the systematic assembly of a nine-marker plasma protein panel spanning all four pathophysiological axes of IS. The signal-to-noise normalisation framework produced technically reproducible measurements across all samples, establishing a clean and biologically interpretable input matrix for downstream modelling.

Aim 2 was fulfilled through the training and evaluation of three supervised classifiers under LOOCV. The ensemble classifiers RF and GB achieved AUC values of 0.722 and 0.724 respectively, substantially outperforming the linear baseline LR at 0.613, confirming that the discriminative structure encoded in the panel is inherently nonlinear. GB's superior sensitivity positions it as the most appropriate architecture for a screening context where missed diagnoses carry the higher consequence.

Aim 3 was fulfilled through permutation-based feature importance analysis conducted within the LOOCV framework, mapping classifier outputs back to the underlying pathophysiological axes of IS. **α2M** emerged as the dominant discriminator across all three classifiers, corroborated by MWU testing and independently supported by published clinical evidence. The secondary contributions of **Fg**, **Fn**, and **NfL**, recoverable through ensemble feature interactions but absent in univariate testing, demonstrated that the diagnostic signal is distributed across multiple pathophysiological axes and is only fully accessible when markers are modelled jointly, directly validating the multi-marker panel design.

**Recommendation & Future work**

The findings presented establish a proof-of-concept foundation for a biologically grounded, multi-biomarker diagnostic framework for IS, but the boundary conditions identified throughout necessarily define a concrete and tractable research agenda. Therefore, the following future work addresses these constraints in order of methodological priority.

The cohort of 30 samples was sufficient for exploratory classification under LOOCV but insufficient for subgroup stratification or a held-out validation split; cohort expansion to at least 150 participants with standardised demographic and clinical metadata is therefore the most pressing requirement. Once a larger dataset is available, a conventional train/test split should replace LOOCV as the primary evaluation framework to reduce the risk of optimistic bias in reported performance estimates.

The fixed nine-marker panel, whilst pathophysiologically motivated, was not subject to systematic optimisation. Recursive feature elimination should be applied not primarily to reduce assay burden, but to definitively establish which biomarkers within each of the four pathophysiological axes contribute most decisively to IS discrimination, providing an empirically grounded basis for targeted panel expansion. Rather than broadening indiscriminately, candidate markers should be recruited to enrich the axes already shown to be diagnostically informative. Within the neuroglial injury and neuroinflammation axis, MMP-9 and S100B are strong candidates given their established elevation in acute IS and demonstrated utility in distinguishing IS from haemorrhagic stroke [32][46]. Within the thrombosis and haemostasis axis, D-dimer and vWF offer complementary signals to the Fg and Fn already included [47][48]. For the systemic inflammation axis, where CRP underperformed, IL-6 represents a higher-specificity alternative with a more favourable temporal profile [49]. Axis-guided expansion of this kind, directed by recursive feature

elimination findings, would transform the panel from a fixed hypothesis-driven design into an iteratively refined, biologically grounded diagnostic instrument. Separately, the reliance on sequential single-plex quantification introduces inter-assay variability that a multiplex platform would substantially mitigate, and adoption of such a platform would be a practical prerequisite for any expanded panel.

Finally, the binary classification framework employed does not reflect the differential diagnostic demands of the emergency setting. Evaluation of the panel against haemorrhagic stroke and transient ischaemic attack (TIA) controls, and validation against an independent multisite cohort, are necessary before the pipeline can be considered for clinical translation. Collectively, these steps represent the methodological pathway from proof-of-concept to clinical translation.

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# Appendix

## 8.1. Appendix A.1 - Exploratory Data Visualisation

```python
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt


# 1. Load dataset
df = pd.read_csv("stroke_heatmap.csv", sep=",")
df.columns = df.columns.str.strip()  # clean column names

# 2. Separate biomarker data and group labels
biomarkers = df.columns.drop("Group")
groups = df["Group"]
data = df[biomarkers]


# 3. Log-transform + global min-max normalization
data_log = np.log1p(data)
min_val = data_log.min().min()
max_val = data_log.max().max()
data_norm = (data_log - min_val) / (max_val - min_val)

# 4. Add group as separate column for annotation
data_norm["Group"] = groups

# 5. Create row color map for Healthy/Stroke
row_colors = data_norm["Group"].map({"Healthy": "green", "Stroke": "red"})

# 6. Plot heatmap
plt.figure(figsize=(len(biomarkers)*0.6, len(df)*0.25), dpi=300)
sns.heatmap(
    data_norm[biomarkers],
    cmap="RdYlGn_r",            # red = high, green = low
    linewidths=0.5,
    yticklabels=groups,
    cbar_kws={"label": "Normalized log(SNR + 1)"}
)
plt.axhline(y=len(data_norm)/2, color="black", linewidth=5)
plt.title("Patient-level Biomarker Heatmap\n(Log-transformed + Global Min-Max Normalized)")
plt.xticks(rotation=45, ha="right")
plt.tight_layout()
plt.show()
```

**Figure 6.** *Python code for generating Figure 3a, the Patient level Biomarker heatmap*

```python
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.read_csv("stroke_heatmap.csv", sep=",")

biomarker_cols = [col for col in df.select_dtypes(include='number').columns if col != 'Label']
corr_matrix = df[biomarker_cols].corr()

plt.figure(figsize=(10, 8))
sns.heatmap(corr_matrix,
            annot=True,
            fmt='.2f',
            cmap='RdYlGn',
            vmin=0,
            vmax=1,
            square=True,
            linewidths=0.5)
plt.title('Biomarker Correlation Matrix')
plt.tight_layout()
plt.savefig('correlation_matrix.png', dpi=300)
plt.show()
```

**Figure 7.** *Python code for computing and visualising Figure 3b, the Pearson correlation matrix*

## 8.2. Appendix A.2 - Machine Learning Classification Pipeline

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.gridspec as gridspec
import seaborn as sns

from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier
from sklearn.model_selection import LeaveOneOut
from sklearn.metrics import confusion_matrix, roc_curve, auc, accuracy_score, recall_score
from sklearn.preprocessing import StandardScaler
from sklearn.inspection import permutation_importance
from sklearn.utils.class_weight import compute_sample_weight

# ── Load data ──────────────────────────────────────────────────────────
df = pd.read_csv('stroke_biomarkers.csv')
X  = df.drop('label', axis=1)
y  = df['label']
feature_names = X.columns.tolist()
N = len(y)

# ── Model registry ─────────────────────────────────────────────────────
# GradBoost does not support class_weight — balanced weighting is passed
# via sample_weight in the fit() call inside run_loocv() instead.
MODEL_REGISTRY = [
    ('Logistic Regression',
     lambda: LogisticRegression(solver='liblinear', max_iter=10_000,
                                class_weight='balanced', random_state=42)),
    ('Random Forest',
     lambda: RandomForestClassifier(n_estimators=100, class_weight='balanced',
                                    random_state=42, n_jobs=-1)),
    ('Gradient Boosting',
     lambda: GradientBoostingClassifier(n_estimators=100, max_depth=3,
                                        learning_rate=0.1, random_state=42)),
]

PALETTE = {
    'Logistic Regression': '#4C72B0',
    'Random Forest':       '#55A868',
    'Gradient Boosting':   '#DDB452',
}
```

```python
# ── Bootstrapped AUC confidence intervals ──────────────────────────────
def bootstrap_auc_ci(y_true, y_probs, n_bootstraps=1000, ci=0.95, random_state=42):
    rng       = np.random.default_rng(random_state)
    y_true    = np.array(y_true)
    y_probs   = np.array(y_probs)
    n         = len(y_true)
    auc_scores = []

    for _ in range(n_bootstraps):
        idx = rng.integers(0, n, size=n)
        # Skip resamples where only one class is present — AUC is undefined
        if len(np.unique(y_true[idx])) < 2:
            continue
        fpr, tpr, _ = roc_curve(y_true[idx], y_probs[idx])
        auc_scores.append(auc(fpr, tpr))

    alpha = (1.0 - ci) / 2.0
    lower = np.percentile(auc_scores, 100 * alpha)
    upper = np.percentile(auc_scores, 100 * (1 - alpha))
    return round(lower, 3), round(upper, 3)

# ── Metrics ────────────────────────────────────────────────────────────
def compute_metrics(res):
    y_true, y_pred, y_probs = res['y_true'], res['y_pred'], res['y_probs']
    fpr, tpr, _  = roc_curve(y_true, y_probs)
    auc_val      = auc(fpr, tpr)
    ci_lower, ci_upper = bootstrap_auc_ci(y_true, y_probs)
    return dict(
        acc      = accuracy_score(y_true, y_pred),
        sens     = recall_score(y_true, y_pred, pos_label=1, zero_division=0),
        spec     = recall_score(y_true, y_pred, pos_label=0, zero_division=0),
        fpr      = fpr, tpr = tpr,
        auc      = auc_val,
        auc_ci   = (ci_lower, ci_upper),   # ← bootstrapped 95% CI
        cm       = confusion_matrix(y_true, y_pred),
    )

# ── LOOCV permutation importance (averaged over folds) ─────────────────
def build_loocv_importance_df(store, feature_names):
    rows = {}
    for name in store:
        arr = store[name]['importances']
        if arr.size > 0:
            vals        = arr.mean(axis=0)
            vals        = np.clip(vals, 0, None)
            rows[name] = vals / (vals.sum() + 1e-8)
    return pd.DataFrame(rows, index=feature_names)
```

```python
# ── LOOCV ──────────────────────────────────────────────────────────────
def run_loocv(X, y, model_registry):

    loo   = LeaveOneOut()
    store = {
        name: {'y_true': [], 'y_pred': [], 'y_probs': [], 'importances': []}
        for name, _ in model_registry
    }

    for fold, (train_idx, test_idx) in enumerate(loo.split(X)):
        X_train, X_test = X.iloc[train_idx], X.iloc[test_idx]
        y_train, y_test = y.iloc[train_idx], y.iloc[test_idx]

        # Fit scaler on train only — prevents data leakage
        scaler      = StandardScaler()
        X_train_sc = scaler.fit_transform(X_train)
        X_test_sc  = scaler.transform(X_test)

        # Balanced sample weights for GradBoost (no class_weight param)
        sw = compute_sample_weight('balanced', y_train)

        for name, factory in model_registry:
            model = factory()

            if name == 'Gradient Boosting':
                model.fit(X_train_sc, y_train, sample_weight=sw)
            else:
                model.fit(X_train_sc, y_train)

            store[name]['y_true'].append(y_test.values[0])
            store[name]['y_pred'].append(model.predict(X_test_sc)[0])
            store[name]['y_probs'].append(model.predict_proba(X_test_sc)[0][1])

            # Permutation importance on training fold (see docstring above)
            perm = permutation_importance(
                model, X_train_sc, y_train,
                n_repeats=10, random_state=42, n_jobs=-1
            )
            store[name]['importances'].append(perm.importances_mean)

        if (fold + 1) % 5 == 0 or (fold + 1) == N:
            print(f"  Fold {fold + 1}/{N} complete …")

    for name in store:
        for k in store[name]:
            store[name][k] = np.array(store[name][k])

    return store
```

```python
# ── Console report ─────────────────────────────────────────────────────
def print_report(metrics_dict, loocv_imp_df, posthoc_imp_df):
    sep = '-' * 90
    print(f'\n{sep}')
    print('  MODEL PERFORMANCE SUMMARY (LOOCV) — ranked by AUC')
    print(sep)
    print(f"  {'Model':<25} {'Acc':>8} {'Sens':>8} {'Spec':>8} {'AUC':>8}  {'95% CI (AUC)'}")
    print(sep)
    for name, m in sorted(metrics_dict.items(), key=lambda x: -x[1]['auc']):
        lo, hi = m['auc_ci']
        print(f"  {name:<25} {m['acc']:>8.3f} {m['sens']:>8.3f} "
              f"{m['spec']:>8.3f} {m['auc']:>8.3f}  [{lo:.3f}, {hi:.3f}]")
    print(sep)

    for label, imp_df in [('LOOCV Training-Fold', loocv_imp_df),
                          ('Post-hoc (Full Data)', posthoc_imp_df)]:
        if imp_df.empty:
            continue
        imp_df      = imp_df.copy()
        imp_df['Mean'] = imp_df.mean(axis=1)
        ranked      = imp_df.sort_values('Mean', ascending=False)
        print(f'\n  FEATURE IMPORTANCE — {label}')
        print(sep)
        print(f"  {'Feature':<20} " +
              "  ".join(f"{c:>22}" for c in ranked.columns))
        print(sep)
        for feat, row in ranked.iterrows():
            vals = "  ".join(f"{v:>22.4f}" for v in row)
            print(f"  {feat:<20} {vals}")
        print(sep)

    print()
```

```python
# ── Master plot ────────────────────────────────────────────────────────
def plot_all(metrics_dict, loocv_imp_df, posthoc_imp_df, model_registry):
    model_names = [n for n, _ in model_registry]
    cmaps       = ['Blues', 'Greens', 'Oranges']

    fig = plt.figure(figsize=(20, 26))
    fig.patch.set_facecolor('#F8F9FA')
    fig.suptitle(
        'Stroke Biomarker Classification — LOOCV Results\n'
        'Logistic Regression  |  Random Forest  |  Gradient Boosting',
        fontsize=15, fontweight='bold', y=0.99
    )

    gs = gridspec.GridSpec(
        5, 3, figure=fig,
        top=0.95, bottom=0.03,
        hspace=0.55, wspace=0.38,
        height_ratios=[1.1, 1.0, 1.1, 1.1, 1.1]
    )

    # ── Row 2: ROC curves ──────────────────────────────────────────────
    ax_roc = fig.add_subplot(gs[2, :])
    for name in model_names:
        m      = metrics_dict[name]
        lo, hi = m['auc_ci']
        ax_roc.plot(m['fpr'], m['tpr'], color=PALETTE[name], lw=2.5,
                    label=f'{name}   AUC={m["auc"]:.3f} (95% CI: {lo:.3f}–{hi:.3f})')
    ax_roc.fill_between([0, 1], [0, 1], [0, 1], alpha=0.06, color='grey')
    ax_roc.plot([0, 1], [0, 1], 'k--', lw=1, alpha=0.4, label='Random chance')
    ax_roc.set(xlim=[0, 1], ylim=[0, 1.02])
    ax_roc.set_title('ROC Curves (LOOCV)', fontsize=11, fontweight='bold')
    ax_roc.legend(loc='lower right', fontsize=9.5, framealpha=0.85)
    ax_roc.set_facecolor('#F8F9FA')
    ax_roc.spines[['top', 'right']].set_visible(False)
    ax_roc.set_xlabel('False Positive Rate', fontsize=10)
    ax_roc.set_ylabel('True Positive Rate', fontsize=10)

    # ── Rows 3 & 4: Feature importance panels ──────────────────────────
    for row_idx, (imp_df, row_title) in enumerate([
        (loocv_imp_df,   'Permutation Importance — LOOCV Training Folds'),
        (posthoc_imp_df, 'Permutation Importance — Post-hoc Full Dataset (cross-check)'),
    ]):
        if imp_df.empty:
            continue
        imp_plot          = imp_df.copy()
        imp_plot['Mean']  = imp_plot.mean(axis=1)
        imp_plot          = imp_plot.sort_values('Mean', ascending=False)

        for col, name in enumerate(model_names):
            if name not in imp_df.columns:
                continue
            ax_imp  = fig.add_subplot(gs[3 + row_idx, col])
            sorted_df = imp_plot[[name]].sort_values(name, ascending=True)
            top3      = sorted_df[name].nlargest(3).index
            bar_colors = ['#FF5700' if f in top3 else PALETTE[name]
                          for f in sorted_df.index]
            bars = ax_imp.barh(sorted_df.index, sorted_df[name],
                               color=bar_colors, edgecolor='white', linewidth=0.5)
            for bar, val in zip(bars, sorted_df[name]):
                ax_imp.text(bar.get_width() + 0.002,
                            bar.get_y() + bar.get_height() / 2,
                            f'{val:.3f}', va='center', fontsize=7.5)
            # Row label on leftmost panel only
            if col == 0:
                ax_imp.set_ylabel(row_title, fontsize=7.5, labelpad=8,
                                  style='italic', color='#444444')
            ax_imp.set_title(f'{name}\nFeature Importance',
                             fontsize=9.5, fontweight='bold')
            ax_imp.set_xlabel('Normalised Importance', fontsize=8)
            ax_imp.tick_params(labelsize=8.5)
            ax_imp.set_facecolor('#F8F9FA')
            ax_imp.spines[['top', 'right']].set_visible(False)
            ax_imp.set_xlim(0, sorted_df[name].max() * 1.22)

    plt.savefig('stroke_final_v2.png', dpi=150, bbox_inches='tight')
    plt.show()
    print("Figure saved → stroke_final_v2.png")
```

```python
# ── Row 0: Confusion matrices ──────────────────────────────────────────
for col, (name, cmap) in enumerate(zip(model_names, cmaps)):
    ax = fig.add_subplot(gs[0, col])
    m  = metrics_dict[name]
    lo, hi = m['auc_ci']
    sns.heatmap(m['cm'], annot=True, fmt='d', cmap=cmap, ax=ax,
                xticklabels=['Pred 0', 'Pred 1'],
                yticklabels=['True 0', 'True 1'],
                cbar=True, linewidths=1, linecolor='white',
                annot_kws={'size': 13, 'weight': 'bold'})
    ax.set_title(f'{name}\n'
                 f'Acc={m["acc"]:.3f}  Sens={m["sens"]:.3f}  Spec={m["spec"]:.3f}\n'
                 f'AUC={m["auc"]:.3f} (95% CI: {lo:.3f}–{hi:.3f})',
                 fontsize=8.5, fontweight='bold', pad=6)
    ax.tick_params(labelsize=7)

# ── Row 1: Grouped metric bar chart ────────────────────────────────────
ax_bar = fig.add_subplot(gs[1, :])
metric_keys   = ['acc', 'sens', 'spec', 'auc']
metric_labels = ['Accuracy', 'Sensitivity', 'Specificity', 'AUC']
x     = np.arange(len(metric_labels))
width = 0.22

for i, name in enumerate(model_names):
    m    = metrics_dict[name]
    vals = [m[k] for k in metric_keys]
    offset = (i - len(model_names) / 2 + 0.5) * width
    bars   = ax_bar.bar(x + offset, vals, width, label=name,
                        color=PALETTE[name], edgecolor='white',
                        linewidth=0.8, alpha=0.92)
    for bar, val in zip(bars, vals):
        ax_bar.text(bar.get_x() + bar.get_width() / 2,
                    bar.get_height() + 0.012,
                    f'{val:.2f}', ha='center', va='bottom',
                    fontsize=8, fontweight='bold')

ax_bar.set_xticks(x)
ax_bar.set_xticklabels(metric_labels, fontsize=11)
ax_bar.set_ylim(0, 1.18)
ax_bar.axhline(0.5, color='grey', linestyle='--', lw=1,
               alpha=0.5, label='Chance (0.5)')
ax_bar.set_title('Performance Metrics Comparison (LOOCV)', fontsize=11, fontweight='bold')
ax_bar.legend(fontsize=9, loc='upper right', framealpha=0.85)
ax_bar.set_ylabel('Score', fontsize=10)
ax_bar.set_facecolor('#F8F9FA')
ax_bar.spines[['top', 'right']].set_visible(False)
```

```python
# ── Main ──────────────────────────────────────────────────────────────────
if __name__ == '__main__':
    print(f"Running LOOCV ({N} folds) — LR, RF, Gradient Boosting …\n")

    store        = run_loocv(X, y, MODEL_REGISTRY)
    metrics_dict = {name: compute_metrics(store[name]) for name, _ in MODEL_REGISTRY}
    loocv_imp_df = build_loocv_importance_df(store, feature_names)

    print("\nComputing post-hoc permutation importance on full dataset …")
    posthoc_imp_df = compute_posthoc_importance(X, y, MODEL_REGISTRY, feature_names)

    print_report(metrics_dict, loocv_imp_df, posthoc_imp_df)
    plot_all(metrics_dict, loocv_imp_df, posthoc_imp_df, MODEL_REGISTRY)
```

**Figure 8.** *Python code for the full machine learning classification pipeline, generating Figures 4a, 4b, 4c, 5a, and 5b*

### 8.3. Appendix A.3 - Statistical Analysis

```python
import pandas as pd
from scipy import stats
import matplotlib.pyplot as plt

# Load your data
df = pd.read_csv('stroke_biomarkers.csv')

# Print columns to verify names
print(df.columns.tolist())

# Separate IS and healthy samples
healthy = df[df['Label'] == 0]
stroke = df[df['Label'] == 1]

# Select biomarker columns only
biomarker_cols = df.select_dtypes(include='number').columns.tolist()
biomarker_cols = [col for col in biomarker_cols if col != 'Label']

# Run Mann-Whitney U test for each biomarker
results = []
for biomarker in biomarker_cols:
    stat, p_value = stats.mannwhitneyu(
        stroke[biomarker],
        healthy[biomarker],
        alternative='two-sided'
    )
    results.append({
        'Biomarker': biomarker,
        'U-statistic': round(stat, 3),
        'p-value': round(p_value, 4),
        'Significant (p<0.05)': 'Yes' if p_value < 0.05 else 'No',
        'Significant (Bonferroni p<0.0056)': 'Yes' if p_value < 0.0056 else 'No'
    })

# Create results dataframe
results_df = pd.DataFrame(results)
print(results_df.to_string(index=False))

# Save table to excel
results_df.to_excel('statistical_results.xlsx', index=False)

# Generate p-value bar chart
fig, ax = plt.subplots(figsize=(10, 6))

# Fixed colors line
colors = []
for p in results_df['p-value']:
    if p < 0.0056:
        colors.append('darkred')
    elif p < 0.05:
        colors.append('coral')
    else:
        colors.append('lightgrey')

ax.bar(results_df['Biomarker'], results_df['p-value'], color=colors,
       edgecolor='black', linewidth=0.5)

ax.axhline(y=0.05, color='coral', linestyle='--',
           linewidth=1.5, label='p = 0.05')
ax.axhline(y=0.0056, color='darkred', linestyle='--',
           linewidth=1.5, label='Bonferroni corrected p = 0.0056')

for i, (biomarker, p_value) in enumerate(zip(results_df['Biomarker'],
                                             results_df['p-value'])):
    ax.text(i, p_value + 0.001, f'{p_value:.4f}',
            ha='center', va='bottom', fontsize=8, rotation=45)

ax.set_xlabel('Biomarker', fontsize=12)
ax.set_ylabel('p-value', fontsize=12)
ax.set_title('Mann-Whitney U Test p-values per Biomarker', fontsize=13)
ax.legend(fontsize=10)

plt.tight_layout()
plt.savefig('mwu_pvalues.png', dpi=300)
plt.show()
```

**Figure 9.** *Python code for the Mann-Whitney U test across all nine biomarkers that generates Figure 5c.*